

ANLP Assignment 3 - Exploring Bias in Lexical and Neural Rankers

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1 Introduction

There has been a revolution in the field of information retrieval with the ever growing emergence of Large Language Models (LLMs). Information retrieval is the extraction of information from a large corpora of unstructured knowledge that satisfies user's needs - for example, web search engines that retrieve relevant web pages based on user query (Wang et al., 2024). The task of information retrieval involves sorting and ranking of documents based on the intent of the user query. Since the field of Information Retrieval involves searching through millions of documents, these documents are typically stored in the form of indexes. Indexes are special data structures that allow easy retrieval and access to documents in search (Kobayashi and Koichi, 2000).

Since LLMs have been pre-trained on large corpora of text, they are not only capable of understanding human language but also capturing the nuances and contextual meaning of the language. With these benefits and increasing adoption of large language models (LLMs), a wide variety of documents on the web are now generated not only by humans but also by LLMs- commonly referred to as Artificial Intelligence Created Content (AIGC) (Dai et al., 2024). As a result, indexed web content increasingly comprises a mix of both human-written and machine-generated text.

This raises an important question: **What is the impact of LLM generated content on neural rankers?** It has been observed that neural retrieval models tend to favor LLM-generated documents over human-written text, leading to a distinct form of bias known as source bias, which manifests not only in first-stage rankers but also persists through second-stage re-rankers (Dai et al., 2024). This literature survey explores the concept of bias in retrieval systems, replicates the findings from the 2024 study 'Neural Retrievers are Biased Towards

LLM-Generated Content' by Dai et al. (2024), and concludes with a proposal to mitigate this bias by introducing mechanisms that offer users greater control over the retrieval process.

2 Problem Statement

LLMs are being widely adopted as a means of generating textual content on the web. Hanley and Durumeric (2024) suggests that the proportion of AIGC has increased by 66% in less than 2 years. With this increasing adoption comes the risk of bias in relation to information retrieval engines. Their study shows that dense retrieval algorithms like BERM (Xu et al., 2023) and ANCE (Xiong et al., 2020) are biased toward LLM generated content.

A similar behavior is observed in Retrieval Augmented Generation (RAG) engines. Originally a concept from communications theory, "Spiral of Silence" (Chen et al., 2024) describes how minority or less visible perspectives tend to disappear from public conversation. This concept has been applied to how LLMs can overshadow human texts in retrieval systems by generating large amounts of content consistently distributed. Experiments have shown that RAG systems with human- and LLM-generated content available to it to search from, consistently prefer LLM-generated content over human-generated content, even when it leads to incorrectness.

Experiments have also shown how AI-generated content (AIGC) has less noise compared to human-generated content and provides lower perplexity scores when LLMs are fed as input to the retrieval models, potentially leading to a systematic preference for such content (Dai et al., 2024). AIGC has shown to take semantic shortcuts to remove this noise, which is naturally present in all human-generated content. Previous studies have also analyzed the preference displayed by embedding models toward different writing styles (Cao, 2025).

Their impact on dense retrieval algorithms in IR engines is not yet explored.

Our study aims to explore several key aspects of this process. First, we propose a method that enables users to control the degree of exposure to AIGC in the final document rankings. Additionally, we will investigate how the choice of embedding models and the writing style of documents influence retrieval performance across a wide range of information domains.

Addressing this problem involves building robust classifiers (e.g., GPTZero, Binoculars, SeqXGPT) that detect and label AI text and then balancing or re-ranking results to reduce overexposure of a single text source.

3 Related Work

The widespread availability of large language models (LLMs) such as ChatGPT, LLaMA, and others has led to an explosion of machine-generated text on the Internet (Hanley and Durumeric, 2024). With a variety of use cases, from writing blog articles, product reviews, letters and educational content, AIGC is set to increase on the web in the future. These models can produce highly coherent language that often passes as human-authored text, contributing to machine-made media. However, a growing body of research suggests that information retrieval (IR) systems may not be fully prepared for this onslaught of AI-generated content (Jiang et al., 2023).

Retriever systems, which find relevant documents for a given user query, form the backbone of modern search engines and retrieval-augmented generation (RAG) pipelines (Guu et al., 2020a). Traditional approaches (e.g., BM25, TF-IDF) rely on keyword matches, while newer “neural” retrievers (e.g., ANCE, Contriever) learn dense embeddings to match queries and documents at a semantic level. Large language models power advanced neural retrievers by encoding both queries and documents into vectors (dense embeddings) in a shared semantic space. The relevance of these embeddings can then be compared using techniques such as cosine similarity. This approach has been especially useful in open-domain question answering, summarization, and chat-based information systems. Moreover, researchers and practitioners increasingly use LLM-based modules as rerankers, or to directly generate final answers in two-stage or multistage pipelines.

Recent studies find that some dense retrieval methods (e.g., Contriever, ANCE) can inadvertently rank LLM-generated text higher than human-written content, even if both contain the same semantic information (Dai et al., 2024). Another study explores how rewriting human-authored content in an “LLM-like” style often leads to higher relevance scores, possibly due to how the embeddings were trained or the stylistic uniformity of AI text (Cao, 2025). This leads to concerns that the system might systematically privilege AI text over human voices. The concerns are twofold:

- Overshadowing human perspectives (sometimes called a “Spiral of Silence,” in which authentic human input recedes when overshadowed by synthetic text)
- Amplifying misinformation or low-quality content if no effective guardrails exist for verifying machine-authored text.

LLM outputs differ by style (formal vs. casual, emotive vs. factual), and these styles in turn can impact how retrieval models score the text. It has also been shown that if queries mimic LLM language patterns, retrieval results might exhibit strong bias toward similarly styled AI-generated text. Approaches like DetectGPT, Binoculars, or MMD focus on differences in log perplexity or log-rank distributions between human and machine text to detect potential AI authorship (Hanley and Durumeric, 2024). A variety of detectors (SeqXGPT, GPTZero, Binoculars) can be integrated with IR systems to label AI vs. human content.

Let’s dive into how different writing styles matter in retrieval systems and to move one step closer towards understanding how to build robust models. Writing style can introduce bias into a system because many text embedding models are trained on data that disproportionately represent certain styles, leading them to favor those styles during inference (Cao, 2024). Recent efforts have focused on creating diverse datasets aimed at improving domain generalization in text embeddings, such as the Massive Text Embedding Benchmark (MTEB). (Muennighoff et al., 2022). Figure 1 shows examples of different writing styles that have been explored with distinct tone and patterns.

In the experiment conducted by Cao (2025), query-document pairs from the Natural Questions (NQ) dataset (Lee et al., 2019) are processed using GPT-4o, which rewrites each document in nine

Style-0: Prompt that says "Please rewrite the following prompt:"	Style-1: Your writing style is formal, efficient, and concise, using professional language and focusing on facts, figures, and data.	Style-2: Your writing style is clear and using simple language, often avoiding idioms or complex sentences.
Style-3: Your writing style is informal, often includes emojis, abbreviations, and internet slang.	Style-4: Your writing style is polite, respectful, and somewhat formal. You use more traditional language and avoid using slang or abbreviations.	Style-5: Your writing style is formal, detailed, and precise manner with structured texts. You use technical language and focus on evidence-based arguments.
Style-6: Your writing style is energetic, motivational, and positive manner.	Style-7: Your writing style is friendly, casual, and empathetic manner with personal anecdotes	Style-8: Your writing style is expressive and emotive (passionate, engaging, empathetic). You use metaphors, analogies, and storytelling to convey your points.

Figure 1: From Cao (2025) that uses different writing styles for the same passage

different styles mentioned in Figure 1. [**]Subsequently, various embedding models—such as LLM2Vec-Llama-2-7b, LLM2Vec-Mistral-7b, and SFR-Embedding-2_R—were applied to each style, resulting in the graphs shown in Figure 2. These graphs illustrate how certain embedding models and writing styles are preferred by rankers, highlighting the presence of bias in retrieval models.

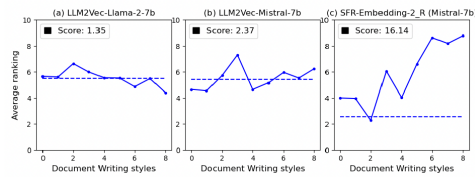


Figure 2: The original document is rewritten in various writing styles and ranked by their similarity to the query using different embedding models. In the resulting graphs, the X-axis represents the different writing styles (as shown in Figure 1), while the Y-axis shows the average ranking of documents in each style for a given embedding model. Each writing style is depicted as a blue dot, and the ranking of the original human-written text is indicated by a dotted line. The relative rankings provide a measure of how biased or unfair a model may be toward specific writing styles. The lower the ranking, higher the preference towards that style (Cao, 2025)

In terms of datasets that have been explored to analyze bias, prior works have tested on NQ320K and SciFact, and have also used WebQ, TriviaQA, and other sets. Research is ongoing on “universal embeddings” such as BGE, GTE, or LLM2Vec for text retrieval and classification, some of which may mitigate bias by training with more balanced data.

Modern retrieval systems such as Retrieval-Augmented Generation (RAG) have recently risen in popularity due to their ability to combine the strengths of information retrieval and generative models, enabling more accurate, grounded, and contextually relevant responses (Guu et al., 2020b; Lewis et al., 2020). In a study by Chen et al. (2024), a simulation experiment was conducted to examine the short-term and long-term effects of gradually introducing LLM-generated content into RAG systems. The findings indicate that, over time, correct answers generated by LLMs gain more prominence compared to correct answers written by humans. Furthermore, the study found that unchecked, erroneous information generated by LLMs was often preferred over correct human-written text, highlighting the potential harm caused by bias in retrieval systems and a need for mitigation of this bias.

4 Experiment

4.1 Baseline Dataset

To answer the research question proposed in the introduction, we are going to use the SciFact+AIGC benchmark introduced in the paper by Dai et al. (2024). The SciFact+AIGC benchmark is constructed using the following methodology: human-written content is first sourced from the original SciFact benchmark, and for each human-authored text, a semantically equivalent LLM-generated counterpart is created using Llama2 (Touvron et al., 2023) and ChatGPT. This pairing enables fair and controlled evaluation of retrieval models across both content sources. This is summarized in the image below by Dai et al. (2024). The original Scifact benchmark (Wadden et al., 2020) was designed to retrieve scientific literature with abstracts for fact checking scientific claims.

4.2 Setup

Our baseline replication experiment is derived from Dai et al. (2024). In this setup, each newly generated LLM-authored document is assigned the same relevance label as its corresponding human-written counterpart, based on the assumption that both convey equivalent semantic content. Formally, $\text{Relevance} \langle \text{query}, \text{human generated document} \rangle$ is equal to $\text{Relevance} \langle \text{query}, \text{LLM generated document} \rangle$. Moreover, the prompt used to create LLM generated content is "Please rewrite the following text: human-written text".

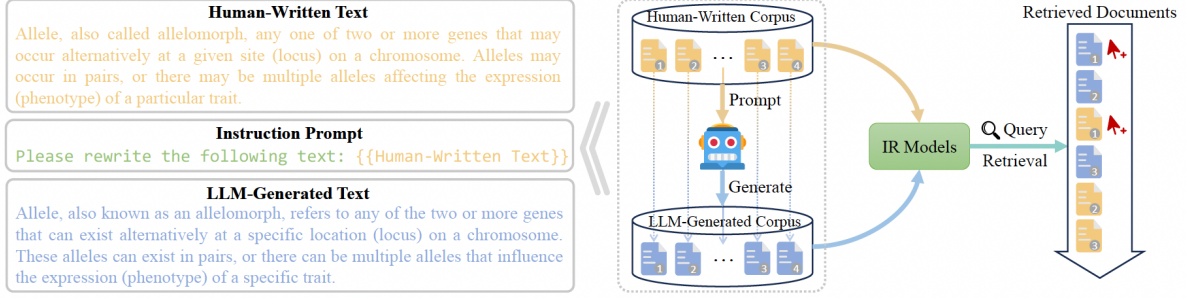


Figure 3: From Dai et al. (2024) Process of creating mix corpora of human and LLM generated documents

With this setup, our experiment aims to address the following questions, which are explored in the subsequent sections.:

1. Do neural rankers exhibit source bias when ranking human-written versus LLM-generated content? If so, to what extent does this bias manifest, and in favor of which type of source?
2. What are the underlying causes of bias in neural rankers? Does this bias propagate through multi-stage retrieval architectures, including second-stage re-rankers?
3. How can the bias in rankers be controlled? Can the user have more control over what kind of results returned by the ranker?

4.3 Evaluation Metrics

To calculate the rankings of both human and LLM generated content, metrics like top K Normalized Discounted Cumulative Gain (NDCG@K) and Mean Average Precision (MAP@K), for K in 1 and 3 is computed separately. When evaluating retrieval performance, documents from alternative sources (e.g. human or LLM generated) are treated as non-relevant for the purpose of metric computation. This allows evaluation of each source separately. Finally, the following expression is used to show the overall bias of the ranker towards either human or LLM generated content. (Dai et al., 2024):

$$\text{Relative}\Delta = \frac{\text{Metric}_{\text{Human-written}} - \text{Metric}_{\text{LLM-generated}}}{\frac{1}{2} (\text{Metric}_{\text{Human-written}} + \text{Metric}_{\text{LLM-generated}})} \times 100\%$$

Tables 1 and 2 show the replicated baseline results for both lexical and neural one staged rankers. Value of "relative metric" above 0 means the retrieval model favors human generated content,

while a value below 0 means the model favors LLM generated content.

5 Project Proposal

5.1 Overview

Our proposal is inspired by the aforementioned experiment and is grounded in the idea of empowering users to curate document rankings by balancing both human-written and AI-generated content (AIGC). This ensures that the results generated match the needs of the user, a step toward reducing bias and promoting fairer information access. Specifically, our approach aims to address the following key objectives:

- **AIGC Exposure Parametrization:** We propose to design a parameter alpha, which ranges from 0 to 1, and governs the exposure that the IR engine has to LLM generated content. The focus of the experiment will be to evaluate the behavior of the IR engine in different settings.
- **Effect on Different Writing Styles:** We aim to see the impact of alpha on the different writing styles of a document.
- **Effect of Embedding Models:** We aim to analyze the effect of different embedding models on the bias shown by an IR engine.
- **Effect of performance with diverse datasets:** We aim to conduct retrieval bias analysis on large and diverse datasets like MSMARCO or TREC to gauge its impact on different types of domain of content in a universal IR scenario.

5.2 Classification method

As a step toward achieving our objective, the first requirement is to accurately distinguish between

Model	Target Corpus	NDCG@1	NDCG@3	MAP@1	MAP@3
ANCE	Human Written	15.3	30.1	14.2	26.2
ANCE	LLM Generated	24.7	35.7	23.3	32.4
ANCE	Relative	-47.0	-17.0	-48.5	-21.2
TAS-B	Human Written	20.0	40.2	19.5	35.2
TAS-B	LLM Generated	31.7	44.8	29.7	41.1
TAS-B	Relative	-45.3	-10.8	-41.5	-15.5
Contriever	Human Written	20.3	40.0	19.3	35.0
Contriever	LLM Generated	33.0	47.0	31.6	43.1
Contriever	Relative	-47.7	-16.1	-48.3	-20.7

Table 1: Performance comparison of neural retrieval models for mixed human written and llama2 generated content on SciFact+AIGC dataset

Model	Target Corpus	NDCG@1	NDCG@3	MAP@1	MAP@3
BM25	Human Written	26.7	40.3	25.7	36.7
BM25	LLM Generated	21.0	38.8	19.6	34.3
BM25	Relative	23.9	3.8	26.9	6.8
TF-IDF	Human Written	22.0	36.9	21.2	33.0
TF-IDF	LLM Generated	17.0	33.8	16.2	29.5
TF-IDF	Relative	25.6	8.8	26.7	11.2

Table 2: Performance comparison of lexical retrieval models for mixed human written and llama2 generated content on SciFact+AIGC dataset

human-written and AIGC. Based on the analysis from Cao (2025), here is a summary of how different do human and AIGC contents look like in Figure 4 using basic statistical analysis on NQ dataset for different kinds of writing styles mentioned in Figure 1.

Figure 4 illustrates how human writing style statistically differs from AIGC. While certain styles—such as Style 0, 1, 2, and 4—exhibit greater similarity to human-written text, others like Style 3, 5, 6, 7 and 9 diverge more significantly. This variability makes it more challenging for a classifier to detect some styles compared to others. We aim to employ a lightweight, practical classifier that leverages the statistical characteristics of both human-written and AIGC, without incurring significant computational overhead.

Accordingly, we adopt the Binoculars classifier in our experiments, as proposed by Hans et al. (2024), due to its ability to achieve state-of-the-art accuracy without requiring any training data. In addition, it can detect AIGC without the need for model-specific adaptations. What makes Binoculars unique is its dual-model approach, which utilizes an ‘observer’ and a ‘performer’ model for detection based on statistical signals. The ‘observer’ model calculates the log perplexity of the docu-

ment, while the ‘performer’ model predicts the next token. These predictions are then used to compute both perplexity and (Hans et al., 2024) cross-perplexity with respect to the ‘observer’ model which is used to classify whether the text is AIGC or human generated.

5.3 Analyzing Embedding Models

The next step in the process is to identify the right embedding model to effectively classify human-generated content and AIGC. BERM and ANCE embeddings Dai et al. (2024) demonstrate the effectiveness of various embedding models and the bias that emerges from the RAG engines that use them. Highlights the applications of universal text embeddings and improvements made in their modern designs.

We plan to use an embedding model that has good fairness scores and also performs well across different writing styles, such as LLM2Vec-Llama-2-7b or mxbai-embed-large-v1.

5.4 Methodology

To implement the framework for analyzing and mitigating bias in IR systems through controlled AIGC exposure, we define a multistage methodology encompassing data preparation, classification,

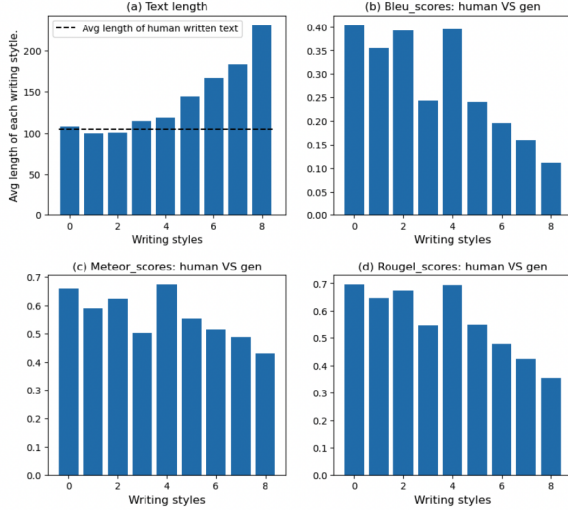


Figure 4: From Cao (2025) showing statistical differences between human generated content vs AIGC in 9 different styles from Figure 1. (a) mentions the average text length of human vs AIGC styles with the dashed line representing average human text length. (b), (c), and (d) are Bleu, Meteor and RougeL scores between human vs AIGC styles.

parameterized exposure, embedding, and evaluation. Here are the steps that are intended to be followed:

1. Dataset preparation: Datasets such as SciFact+AIGC, MSMARCO, and TREC will be used to curate both human-written and AI-generated content. AI-generated documents will be appended to their corresponding human-written counterparts using various writing styles illustrated in Figure 1. This approach not only accounts for the diversity of AIGC likely to appear online but also helps construct a dataset that more accurately reflects real-world variations in writing style.
2. To evaluate the accuracy of the classifier, we will construct a test set with ground truth labels, consisting of documents written in various styles from Figure 1 and encoded using fair embedding models. We will then apply the zero-shot Binoculars classifier to assess its robustness in distinguishing between human-written and AI-generated content. In this way, we can gain insights into the classifier’s performance across diverse writing styles, which is crucial for understanding its generalizability and reliability in the next stage of experiment.
3. Finally, different alpha values for the exposure

parameter will be chosen to determine to what extent AIGC content will be included in the final rankings. New rankings are created by filtering all documents with AIGC-detected content based on the threshold alpha value. This algorithm is defined in detail in the De-bias algorithm subsection.

4. Lastly, experiments will be conducted to see how Relative Δ value changes with change in alpha. Metrics that will be used in the evaluation are: MAP@1, MAP@3, NDCG@1 and NDCG@3. Our hypothesis is that smaller the alpha value, higher the value of Relative Δ since small proportion of AIGC documents are expected to be in the final rankings.

5.5 Debiased algorithm pseudocode

The pseudocode to rank documents in a debiased manner is inspired by xQuAD (Santos et al., 2010). Instead of diversity intent scores, the probability that a document is AIGC as judged by our classifier is attached as metadata to all document chunks.

Algorithm 1 AIGC-Aware Debiased Ranking

Require: Initial ranked list of documents $R = \{d_1, d_2, \dots, d_n\}$
Require: Desired ranking length τ
Require: Exposure control parameter $\alpha \in [0, 1]$
Require: Each document d has: relevance score $\text{rel}(d)$ and AIGC probability $\text{llm}(d)$
Ensure: Diversified ranking S of length τ

```

1:  $S \leftarrow \emptyset$   $\triangleright$  Initialize final ranking
2:  $R' \leftarrow R$   $\triangleright$  Working copy of initial ranking
3: while  $|S| < \tau$  do
4:    $\text{max\_score} \leftarrow -\infty$ 
5:    $\text{selected\_doc} \leftarrow \text{None}$ 
6:   for all  $d \in R'$  do
7:      $\text{human}(d) \leftarrow 1 - \text{llm}(d)$   $\triangleright$  Compute human-authorship probability
8:      $\text{div}(d) \leftarrow \alpha \cdot \text{llm}(d) + (1 - \alpha) \cdot \text{human}(d)$ 
9:      $\text{score}(d) \leftarrow (1 - \alpha) \cdot \text{rel}(d) + \alpha \cdot \text{div}(d)$ 
10:    if  $\text{score}(d) > \text{max\_score}$  then
11:       $\text{max\_score} \leftarrow \text{score}(d)$ 
12:       $\text{selected\_doc} \leftarrow d$ 
13:    end if
14:  end for
15:   $S \leftarrow S \cup \{\text{selected\_doc}\}$ 
16:   $R' \leftarrow R' \setminus \{\text{selected\_doc}\}$ 
17: end while
18: return  $S$ 

```

Using this AIGC probability score, the relevance of the document to the query as judged by the embedding model, and the user-defined value of the exposure control parameter, the algorithm generates a ranking of the documents.

6 Contributions and Acknowledgments

- **Member 1:** Aditi Saini
Contributions: Finding relevant research papers, Getting the evaluation of retrieval models to work for baseline results, Analysis of limitations of the current retrieval models in rankings, Ideation of the debias algorithm for the project proposal, and Report writing
- **Member 2:** Aditya Shrivastava
Contributions: Finding relevant research papers, Working on debugging the baseline results for the debias algorithm, Implementaion of the pseudocode for the debias algorithm, Research into different embedding models, and Report writing .

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